

Progress Report

Link Structure & Knowledge Flow Dashboard (Wikipedia Internal Link Analysis)

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1 Project Objectives

Design and implement a reproducible analytics and visualisation system that models Wikipedia's internal link structure as a dynamic graph, with the goal of understanding how knowledge is organised, navigated, and redistributed within a chosen topic over time.

1.1 Steps Taken:

1. Link ingestion
2. Graph Construction
3. Graph analysis/ mathematics
4. Interactive visualisation

2 Major Algorithms

2.1 Temporal Analysis:

To analyse structural evolution of the Wikipedia link network, we compare edge sets across time snapshots.

If,

$A(t_i)$ = edge set at time t_i

$A(t_{i-1})$ = edge set at previous snapshot

Then:

Links added: $|A(t_i) \setminus A(t_{i-1})|$

Links removed: $|A(t_{i-1}) \setminus A(t_i)|$

This helps identify: Growth of knowledge areas, Emerging topics, Structural rewiring, Stability of important pages

2.2 Structural Shifts:

Louvian algorithm:

Helps detect communities of closely related articles. The algorithm maximizes modularity, which measures how densely connected nodes are within a group compared to random connections.

Applications in this project:

- Group related Wikipedia topics
- Detect categorical segregation
- Track evolution of topic groups over time and understand how knowledge naturally organizes itself

Strongly connected components:

Algorithms used: Kosaraju or Tarjan algorithm

Applications:

- Identify tightly related topic clusters
- Study knowledge cycles
- Measure maximum SCC size to understand topic cohesion

Centrality measures:

- Degree centrality- Measures importance based on connectivity.
- Betweenness centrality- Measures how often a node lies on shortest paths.

BFS:

- Finding the shortest distance between two nodes

3 Current Status of Work

Code to link extraction and graph construction

- Pratik (Basic working prototype)
- Aadya (code for constructing a graph using different revision IDs, checking basic graph measures)
- Anshita (graph temporal analysis)
- Nidhi (code for constructing a graph with multiple hops, making a time consistent graphs for previous revisions)

4 Team Work Distribution

- Understanding basics of link extraction and graphing - All
- Graph Construction, Data Storage Methods - Nidhi Bhagwat
- Graph Mathematics and Analysis: – Centrality measures - Pratik Sahoo
 – Structural Shifts - Anshita Sharma
- Interactive Dashboard - Aadya Ashwin

5 Expected Outcome

- An interactive dashboard that allows the user to visualise how data is stored in Wikipedia with sliders for seeing temporal shifts in data organisation.
- Visually understand graph measures like centrality, connectivity, etc.
- If time permits, finding the shortest distance between two articles (our version of the wiki game).